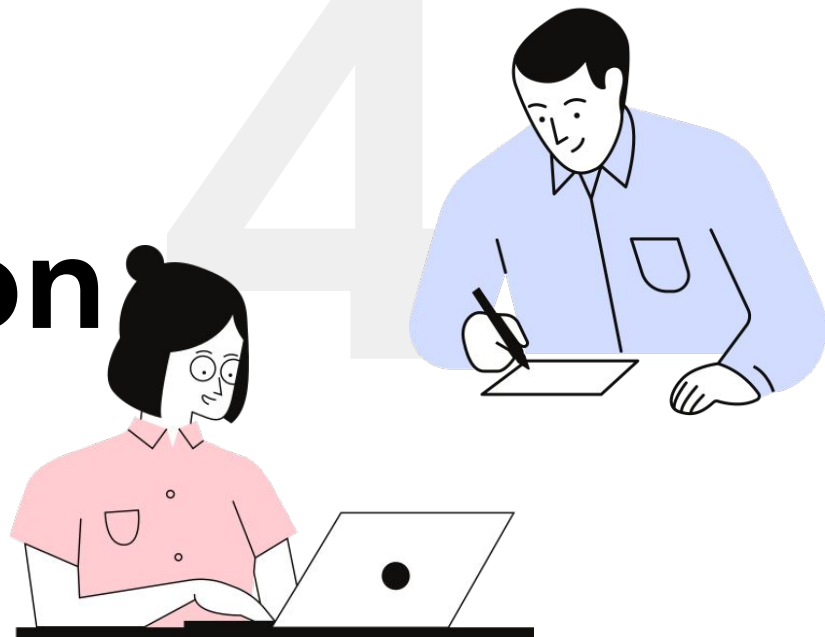




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Information & Communication Technology



Networks

- ❖ Devices: Routers, NICs, hubs, switches, and bridges.
- ❖ Wireless: Wi-Fi (fast, long-range) and Bluetooth (short-range, low-bandwidth).
- ❖ Cloud Computing: Public, private, and hybrid storage systems.
- ❖ Environments: Intranets, extranets, and the internet.
- ❖ Network Types: LANs, WLANs, and WANs.

Network Issues and Communication

- ❖ Security: Passwords, authentication (biometrics, tokens, etc.).
- ❖ Anti-Malware: Virus protection with anti-virus software.
- ❖ Conferencing:
 - Video: Real-time video and sound.
 - Audio: Phone or VoIP-based meetings.
 - Web: Internet-based with collaboration tools.

Network Interface Card (NIC)

- ❖ Enables a device to connect to a network.
- ❖ Converts binary data into electrical signals for network access.
- ❖ Integrated into the motherboard of modern computers.

Media Access Control (MAC) Address:

- ❖ Unique hard-coded identifier for the device, assigned during manufacturing.

Wireless NIC (WNIC):

- ❖ Connects devices via wireless networks using microwaves.
- ❖ Often a USB plug-in or integrated circuit.

Media Access Control (MAC) Address

- ❖ A unique 48-bit identifier for devices on a network.
- ❖ Format: Six hexadecimal digit groups (e.g., 00-1C-B3-4F-25-FF).
- ❖ First six digits: Manufacturer's code.
- ❖ Last six digits: Device's serial number.

Key Features:

- ❖ Static and unchanging (even if device location changes).
- ❖ Useful for identifying network faults and ensuring reliable communication.
- ❖ Also Known As: Physical address.

00 – 1C – B3 – 4F – 25 – FF

NN – NN – NN – DD – DD – DD
manufacturer's code device serial number

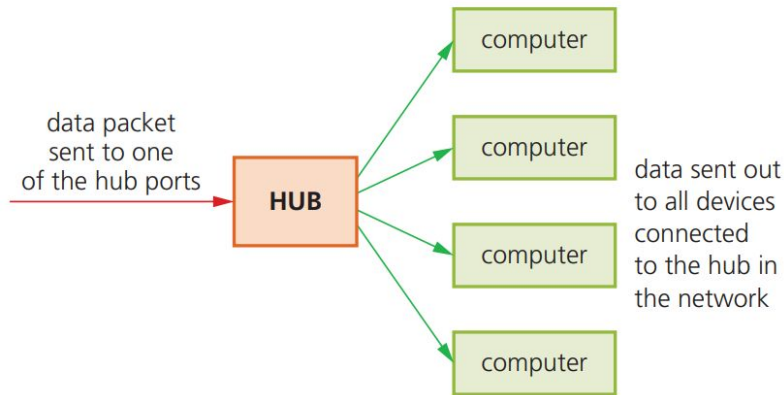
Internet Protocol (IP) Address

- ❖ A unique address assigned to a device when it connects to the internet.
- ❖ Assigned By: Internet Service Provider (ISP).
- ❖ Identifies a device's location on the network.
- ❖ Characteristics:
 - Changes when the device connects from a different location.
 - Unlike MAC addresses, IP addresses are dynamic.
- ❖ Versions:
- ❖ IPv4: 32-bit address (e.g., 254.25.28.77).
- ❖ IPv6: 128-bit address for more devices and traffic (e.g., A8FB:7A88:FFF0:0FFF:3D21:2085:66FB:F0FA).

- ❖ Units of data transmission over a network.
- ❖ Components of Packet Headers:
 - Sender's and receiver's IP addresses.
 - Sequence/identity number (for reassembly).
 - Packet size (for integrity verification).
 - Total number of packets in the message.
- ❖ Routing:
 - Packets pass through multiple routers before reaching their destination.
 - Routers determine the next path using stored routing tables.
 - Assembly: Reassembled in the correct order at the receiving station.

Hubs

- ❖ Definition: Hardware devices that connect multiple devices in a local area network (LAN).
- ❖ Receives data packets at one port and broadcasts them to all connected devices.
- ❖ Advantages:
 - Simple network setup within a building.
- ❖ Disadvantages:
 - Security Risk: All devices receive every data packet.
 - Inefficiency: Unnecessary network traffic reduces bandwidth.



Switches

- ❖ Intelligent devices connecting multiple devices to form a LAN.
- ❖ Stores MAC addresses of all devices in a look-up table.
- ❖ Each port corresponds to a device's MAC address.
- ❖ Matches incoming data packet's MAC address to the look-up table and sends it only to the intended device.
- ❖ Reduces unnecessary network traffic and increases security.
- ❖ Advantages:
 - More secure: Only the intended device receives data.
 - Efficient: Reduces bandwidth usage by avoiding unnecessary data distribution.



Switches cont.

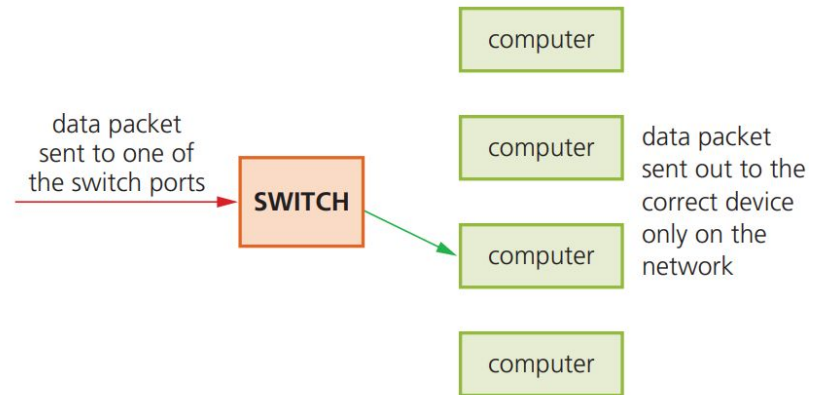
❖ Limitations:

- Only operates within a LAN; cannot connect to external networks like the internet.

❖ Comparison with Hubs:

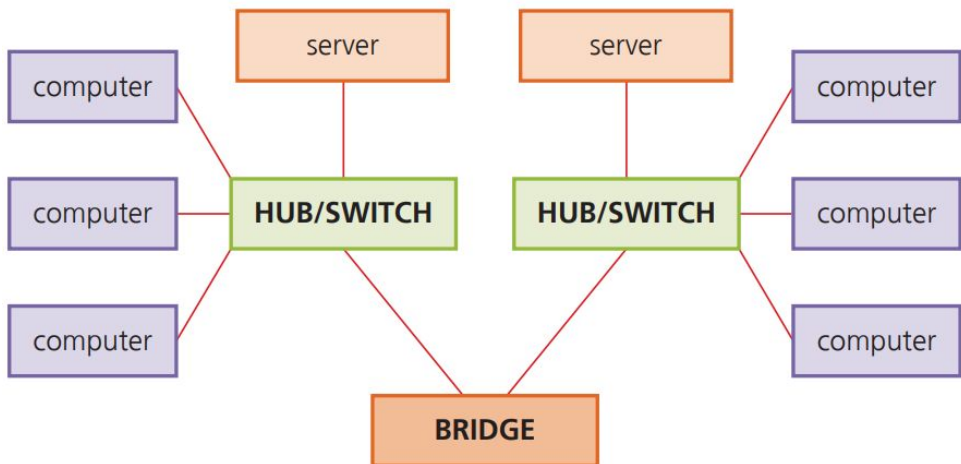
- Hubs send data packets to all connected devices, while switches direct packets to specific devices.
- Switches use MAC addresses for locating the destination device.

Port number	MAC address
1	a4-00-22-a4-fe-d1
2	00-1c-b3-4f-25-ff
3	33-11-ad-6f-f1-00
4	a4-00-22-b2-24-11
5	00-1c-b3-44-ff-02
6	0d-3e-4f-1a-22-00

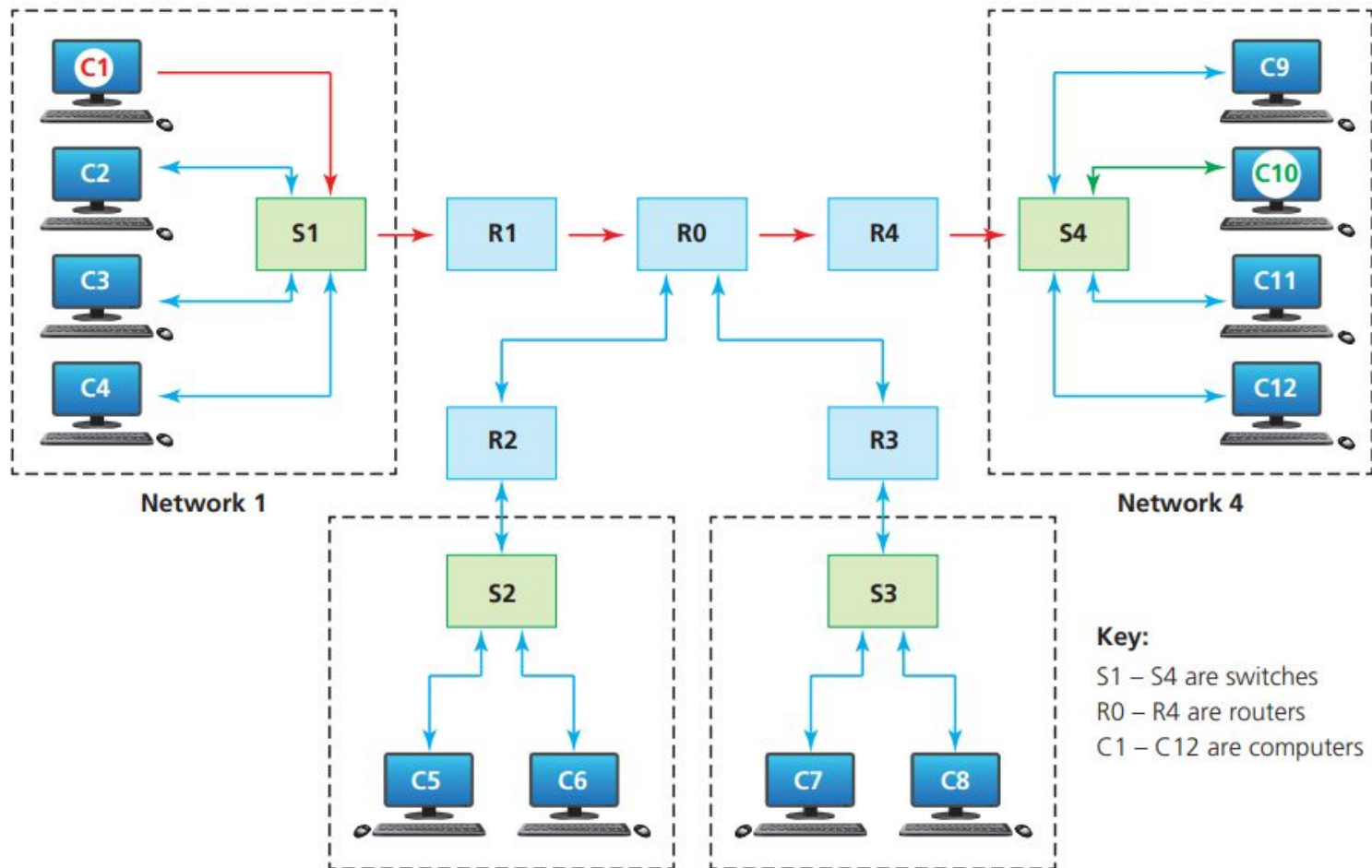


Bridges

- ❖ Devices connecting two LANs using the same communication protocol.
- ❖ Connects different parts of a LAN to function as a single network.
- ❖ Cannot communicate with external networks like the internet.
- ❖ Use Case: Extending or combining LANs.



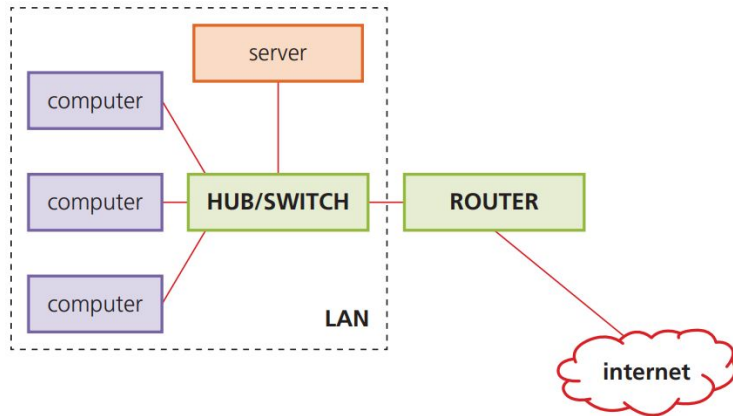
- ❖ Devices that route data packets between networks based on IP addresses.
- ❖ Connects a LAN to the internet.
- ❖ Inspects incoming data packets and determines whether they are for its own network or an external network.
- ❖ Uses routing tables stored in RAM to decide packet destinations.
- ❖ Forwards packets to other routers or internal switches/hubs.
- ❖ Differences from Switches:
 - Routers use IP addresses for routing; switches use MAC addresses.
 - Routers enable communication between different networks, while switches operate within a single LAN.
- ❖ Modern Routers often combine the functionality of routers and switches, storing both MAC and IP addresses for efficiency.



▲ **Figure 4.9** Routing of data from **C1** to **C10**

Comparison of Routers and Bridges

Routers:	Bridges:
Connect various types of networks together.	Connect LANs together.
Scan a device's IP address.	Scan a device's MAC address.
Use data packets for communication.	Use data packets for communication.
Can connect networks with different protocols.	Only connect networks with the same protocols.
Use routing tables to direct data packets.	Do not use routing tables.
Have more than two ports.	Have only two ports.



Bluetooth

- Uses 79 frequency channels centered at 2.45 GHz.
- Employs spread-spectrum frequency hopping to minimize interference.
- Security: Key encryption creates a secure Wireless Personal Area Network (WPAN).
- ❖ Best Use Cases:
 - Close-range data transfer (<30 meters).
 - Low-bandwidth applications (e.g., transferring music files).
 - Non-critical data transmission speeds.
- ❖ Key Features:
 - Maximum data transfer rate: 25 Mbits/second.
 - Can connect up to 7 devices.

Wi-Fi

- Operates on multiple frequency bands (e.g., 2.4 GHz, 5 GHz).
- The 5 GHz band offers faster data transfer but shorter range.
- Security: Uses WEP and WPA for data transmission security.

❖ Best Use Cases:

- Full-scale networks (e.g., home internet).
- Faster data transfer rates.
- Better range (up to 100 meters, reduced by obstructions like walls).

❖ Key Features:

- Maximum data transfer rate: 250 Mbits/second.
- Number of connected devices depends on the router.

Comparison of Wi-Fi & Bluetooth

Feature	Wi-Fi	Bluetooth
Transmission Frequency	2.4, 3.6, 5.0 GHz	2.4GHz
Data Transfer Rate	250 Mbits/second (~31 Mbytes/second)	25 Mbits/second (~3.1 Mbytes/second)
Range	100m	30m
Maximum Devices	Varies	Up to 7
Security	WEP & WPA	Key Matching Encryption

Cloud Storage

- ❖ Data storage on remote servers managed by hosting companies.
- ❖ Offers data redundancy: Data stored on multiple servers for accessibility during maintenance or repair.

- ❖ Types of Cloud Storage:
 - Public Cloud: separate entities for customer and provider.
 - Private Cloud: dedicated environment within a company firewall.
 - Hybrid Cloud: combines public and private clouds for sensitive and less-sensitive data.

Cloud & Disadvantages

Advantages

- Access files anytime, anywhere with internet access.
- No need for external storage devices.
- Remote backup protects against data loss or hardware failure.
- Unlimited storage capacity (at a price).
- Synchronization across devices ensures up-to-date files.
- Ideal for collaboration on shared documents.

Disadvantages

- Security Risks: Data stored externally may be vulnerable to breaches or misuse.
- Internet Dependency: Unstable connections hinder data access.
- Costs: High costs for large storage capacities or high data transfer rates.
- Provider Risks: Cloud company failure could lead to data loss.

Data Security Concerns

- Physical security of data centers.
- Resistance to natural disasters or power outages.
- Safeguards against insider threats from cloud service employees.

Data Loss Risks

- Vulnerable to hacking or pharming attacks.
- 2019 Cultura Colectiva Breach: exposed 540 million Facebook accounts on a cloud server, including sensitive user data.

Local Area Network (LAN)

- ❖ Network within one building or geographically close locations.
- ❖ Components:
 - Computers, printers, and devices connected to hubs or switches.
 - Router connects the LAN to external networks like the internet.

LAN Advantages & Disadvantages

Advantages

- Shared resources (e.g., printers, software).
- Easy communication (e.g., text messaging).
- Managed by a network administrator (monitors security and data traffic).

Disadvantages

- Easier spread of viruses across the network.
- Queues for shared resources (e.g., printers).
- Slower access to external networks.
- Increased security risks compared to stand-alone computers.
- Dependence on the main server for functionality.

Wireless Local Area Network (WLAN)

- ❖ LAN with wireless communication via radio or infrared signals.

Components

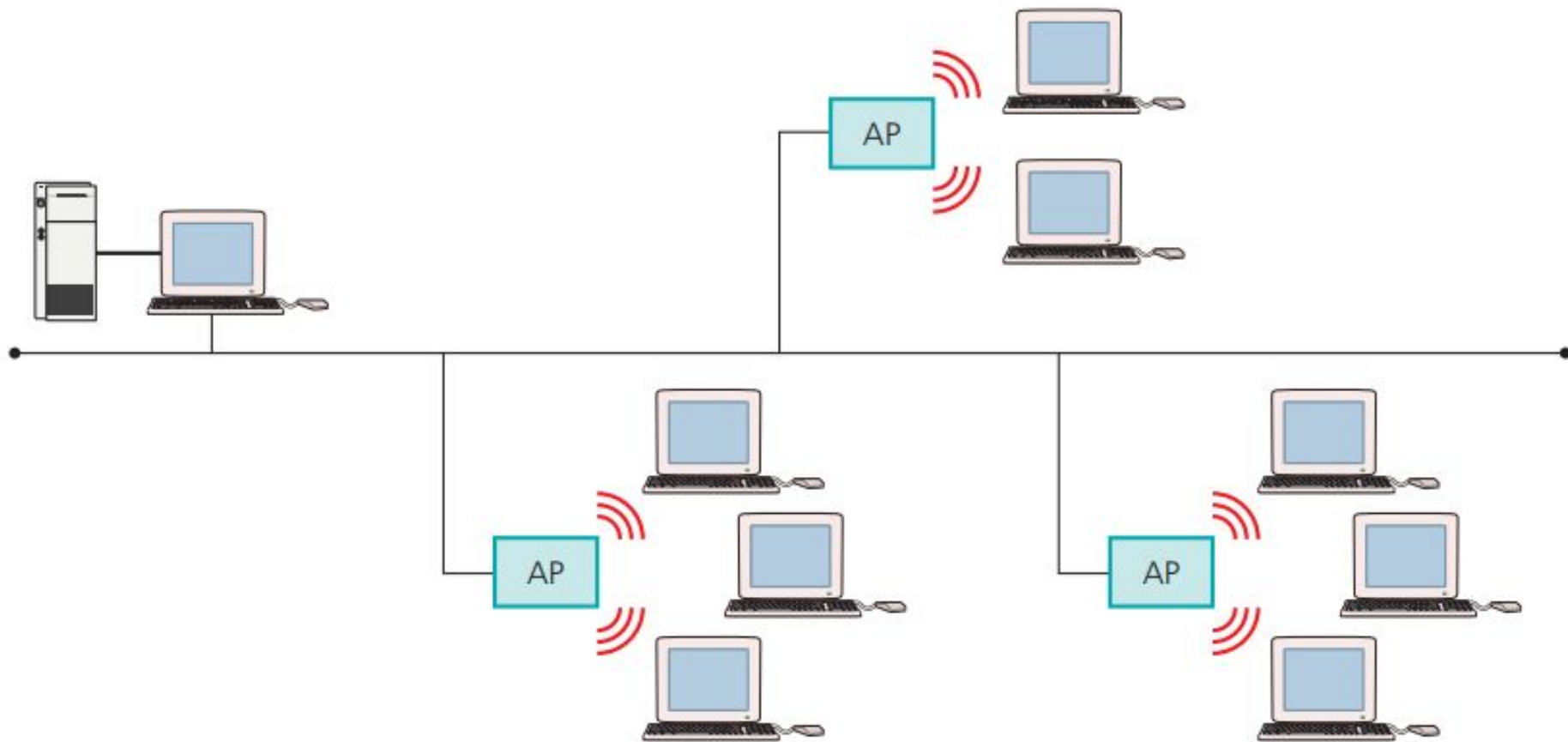
- Access points (APs) connected to a wired network.
- Wireless LAN adapters in devices.

Advantages

- No cables (improved safety and flexibility).
- Increased mobility within the AP range.

Disadvantages

- Susceptible to interference and signal obstruction.
- Slower data transfer rates compared to wired LANs.
- Security vulnerabilities (radio wave interception).



Comparison: Wired vs. Wireless LANs

Wireless Networking

- Easier expansion.
- Increased mobility within AP range.
- No cables (improved safety).
- Slower data transfer and higher security risks.

Wired Networking

- More reliable and stable.
- Faster data transfer rates.
- Cheaper overall but less flexible (requires proximity to cables).
- No signal interference but prone to physical hazards (e.g., tripping, overheating).

- ❖ Network spanning long geographic distances (e.g., between cities or countries).
 - Examples: Bank ATM networks.

Characteristics

- Formed by connecting LANs with routers.
- Utilizes public networks (e.g., telephone lines, satellites) or leased communication lines.
- More secure with dedicated lines.

Size Guide

- WAN: 100 km to >1000 km.
- MAN (Metropolitan Area Network): 1 km to 100 km (for comparison only, not in syllabus).
- LAN: 10 m to 1000 m (1 km).

4.2.1 Security Issues Regarding Data Transfer

- Internet security issues: Hacking, phishing, pharming, and viruses (detailed in Chapter 8, Section 8.3).

4.2.2 Passwords

- Used for: accessing email, online banking, social networks, etc.

Password Protection Methods

- Use anti-spyware software to prevent password theft.
- Regularly change passwords to prevent unauthorized access.
- Use strong passwords: at least one uppercase letter, one number, and one special character.

Zero Login and Biometrics

- Relies on smart devices recognizing users via biometrics and behavioral patterns.
- Builds user profiles for authentication (e.g., fingerprint or face recognition).

Advantages

- Enhanced security (difficult to replicate biometrics).
- Faster and easier login process.

Disadvantages

- Monitoring and logout processes may be unclear.
- Questions about actual protection and privacy.

Magnetic Stripe Cards

- Contain binary data stored in tiny magnetic particles.
- Used for security purposes (e.g., access cards).
- Often include holograms or photos to prevent forgery.

Advantages

- Easy to use and cost-effective.
- Can be remotely deactivated.
- Multi-purpose (e.g., vending machines, door access).

Disadvantages

- Less secure (data can be copied).
- Prone to wear and reader malfunctions.

Smart Cards

- Embedded with RFID tags for contactless use.
- Stores user data (e.g., name, PIN, date of birth).
- Access granted after verifying details via a keypad.

Advantages

- Convenient and secure.
- Read from a distance, no need to swipe.
- Use Case: Security gates, banking, network access.

Physical Tokens

- Solid objects used as authentication tools.

Functionality

- Generates a one-time password (OTP) valid for less than a minute.
- Enhances security by requiring the token, PIN, and card.

Example (Banking)

- Insert debit card into token device (authentication step 1).
- Enter PIN on the device (authentication step 2).
- Enter OTP shown on the device into the banking website

Types of Physical Tokens

- **Disconnected** Tokens: separate devices requiring manual data entry using a keypad.
- **Connected** Tokens: transmit one-time passwords (OTP) directly to a computer via USB.

Electronic Tokens

- Software installed on a user's device (e.g., smartphone).
- Generates an OTP for authentication (valid for less than a minute).
- User enters the OTP with another authentication method (e.g., PIN, Touch).
- Server and app use synchronized clocks to verify OTPs.

Advantages

- Enhances security.
- Eliminates the need for physical devices.

Focus: protecting devices against potential virus attacks with anti-virus software.

- Scans software and files before they are run.
- Compares potential viruses against a database of known viruses.

Heuristic Checking

- Detects suspicious behavior in software not in the virus database.
- Quarantines infected files or programs:
- Automatically deletes or allows user decision.

Key Features

- Needs regular updates due to new virus discoveries.
- Full system scans (e.g., weekly) to detect dormant viruses.

Video Conferencing

- Combines video and sound for real-time communication.
- Hardware: Webcams, microphones, monitors, speakers.
- Software: Drivers, CODEC, echo cancellation.

Advantages

- Quick access to documents or experts.
- Cost-effective (saves on travel and accommodation).
- Environmentally friendly (reduces travel pollution).
- Connects remote employees to headquarters.
- Allows for short-notice conferences.

Disadvantages

- Time lag or jerky images due to network issues.
- High setup and training costs.
- Relies on a stable internet connection.
- May demotivate employees who value international travel.

Audio Conferencing

- Uses sound-only communication.
- Simpler and less resource-intensive than video conferencing.
- Meetings held using audio equipment via telephone or Voice over Internet Protocol (VoIP).

4.2.5 Electronic Conferencing

Procedure for Phone Conference

- Organiser receives two PINs:
 - Personal PIN for organiser (e.g., 2151).
 - Participant PIN for attendees (e.g., 8422).
- Organiser shares PIN, date, and time with participants.
- At the scheduled time:
 - Organiser dials the conference number and enters their PIN.
 - Participants dial the same number and input their PIN to join.

Equipment

- Standard telephone for phone conferencing.
- VoIP setup: Computer (with microphone and speakers) or internet phone.

VoIP Features

- Allows voice, instant messaging, and video communication.
- Groups created by dragging and dropping user details.
- Adds landline or mobile numbers for non-internet participants.

Drawbacks

- Sound quality issues like "drop out," echoing, or noisy lines due to unstable broadband.

Web Conferencing

- ❖ Internet-based platform combining text, audio, and video.
- ❖ Suitable for presentations, training, or large virtual meetings.
- ❖ Real-time conferencing over the internet (also called webinars or webcasts).
- ❖ Combines features of audio and video conferencing.
- ❖ Supported on tablets, smartphones, and other devices with webcams and built-in microphones.

Requirements

- Computer with high-speed, stable internet.
- Conference application or link provided by the organiser.

Features

- ❖ Presentations posted in advance, live screen sharing for demonstrations
- ❖ Interactive whiteboards for drawing/writing, document sharing by uploading files to the conference website.
- ❖ Instant messaging and verbal communication, webcam usage for transmitting videos/images.
- ❖ Organiser decides who speaks delegates can raise a virtual flag to request to speak